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Privacy 4FMs

April 23rd, 2026

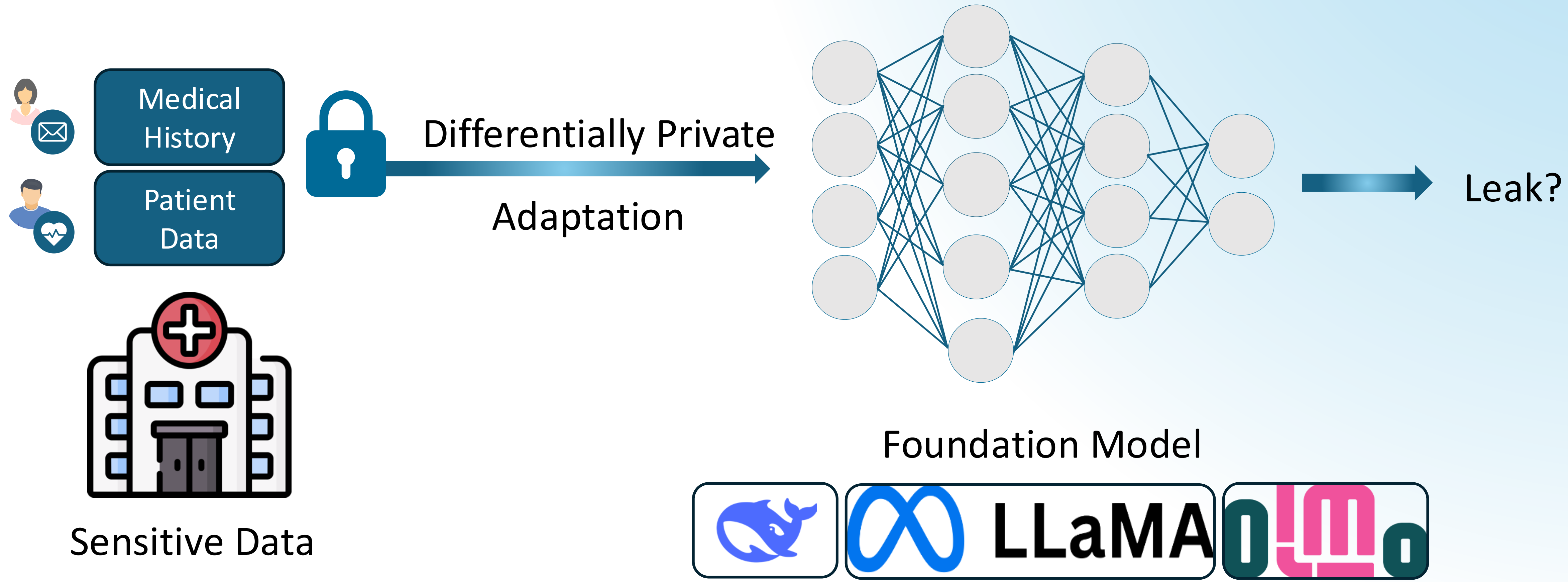
Benchmarking Empirical Privacy Protection for Adaptations of Large Language Models

Franziska Boenisch & Adam Dziedzic

ICLR 2026, Oral

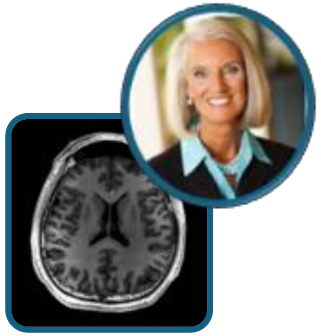
Motivation for Our Work

Currently: no understanding of the actual privacy implications for adaptations



FMs Put Individuals' Privacy at Risk

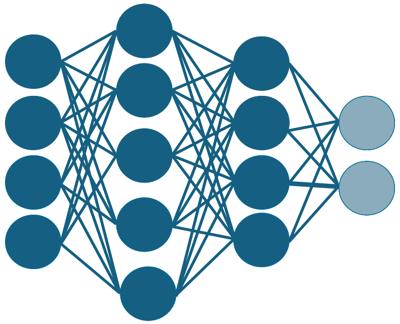
Training



1. Pretraining

Differential Privacy (DP)

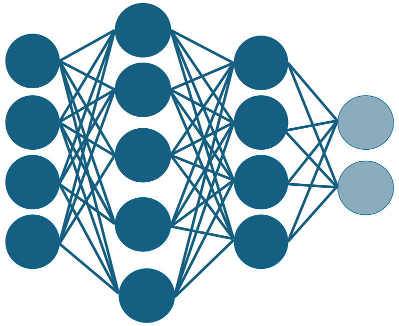
2. Adaptation



DP- Full fine-tuning
DP- Low rank adapters
DP- Prompting

Inference

Query

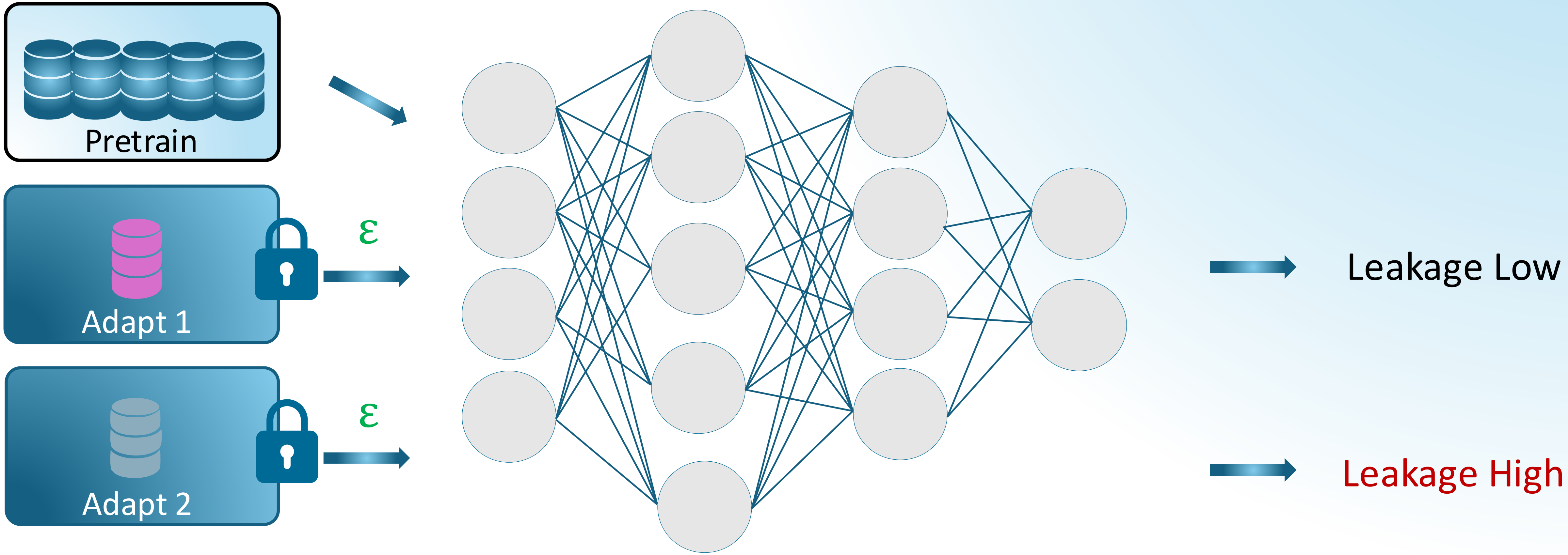


Leak

Bounded by ε

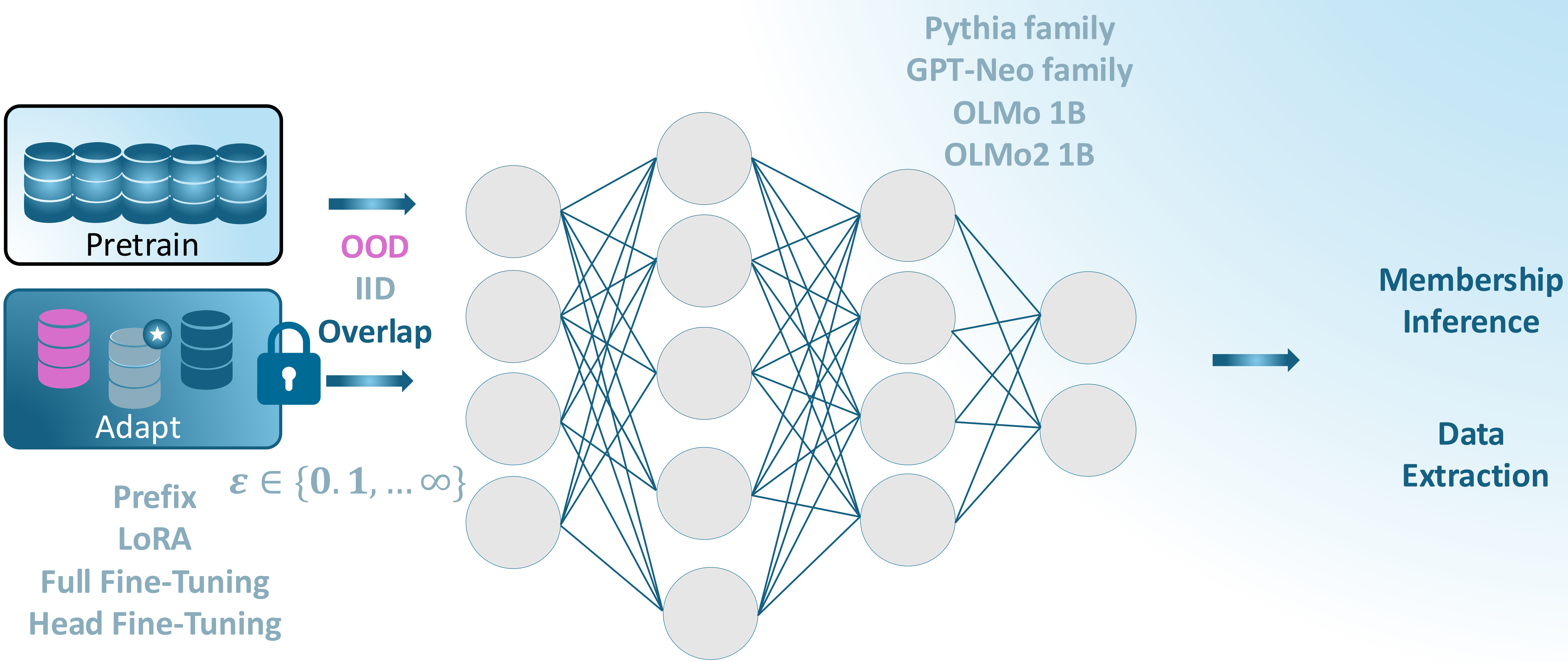
The Key Finding for DP Adaptations

Practical data leakage differs for the same adaptation guarantees



Our Benchmark

What empirical privacy risk remains after DP adaptations?



01 Data Distribution Influences Leakage

The closer the adaptation data to pretraining, the more it leaks, even with no overlap

	OOD (German Wiki)	IID (Books Val)	Overlap (Books Train)
Prefix Tuning			
LoRA			
Full Fine-Tuning			
Head Fine-Tuning			

AUC, $\varepsilon = 8$, Pythia 1B, pretrained on the Pile, including Books Train (Bookcorpus2 Train)

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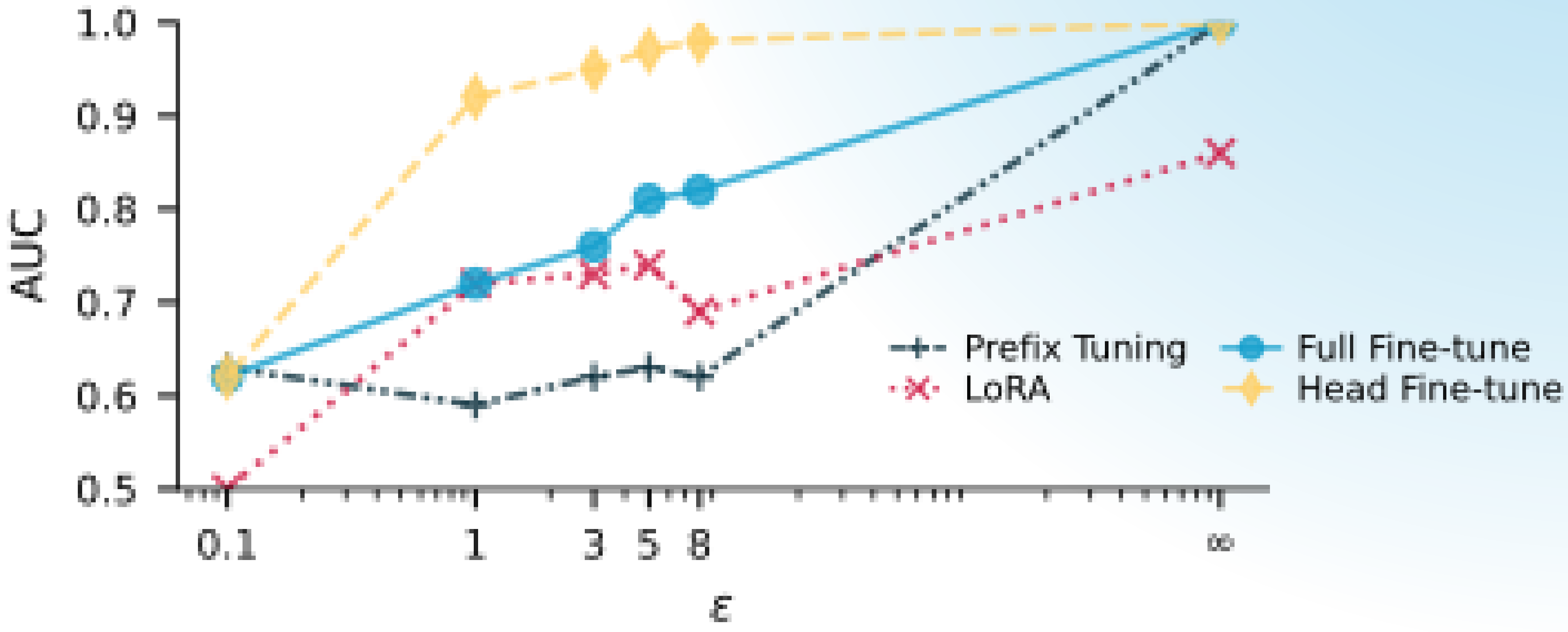
02 Same ε, Different Privacy Leakage

At the same ε and similar utility, leakage from adaptations differs substantially.



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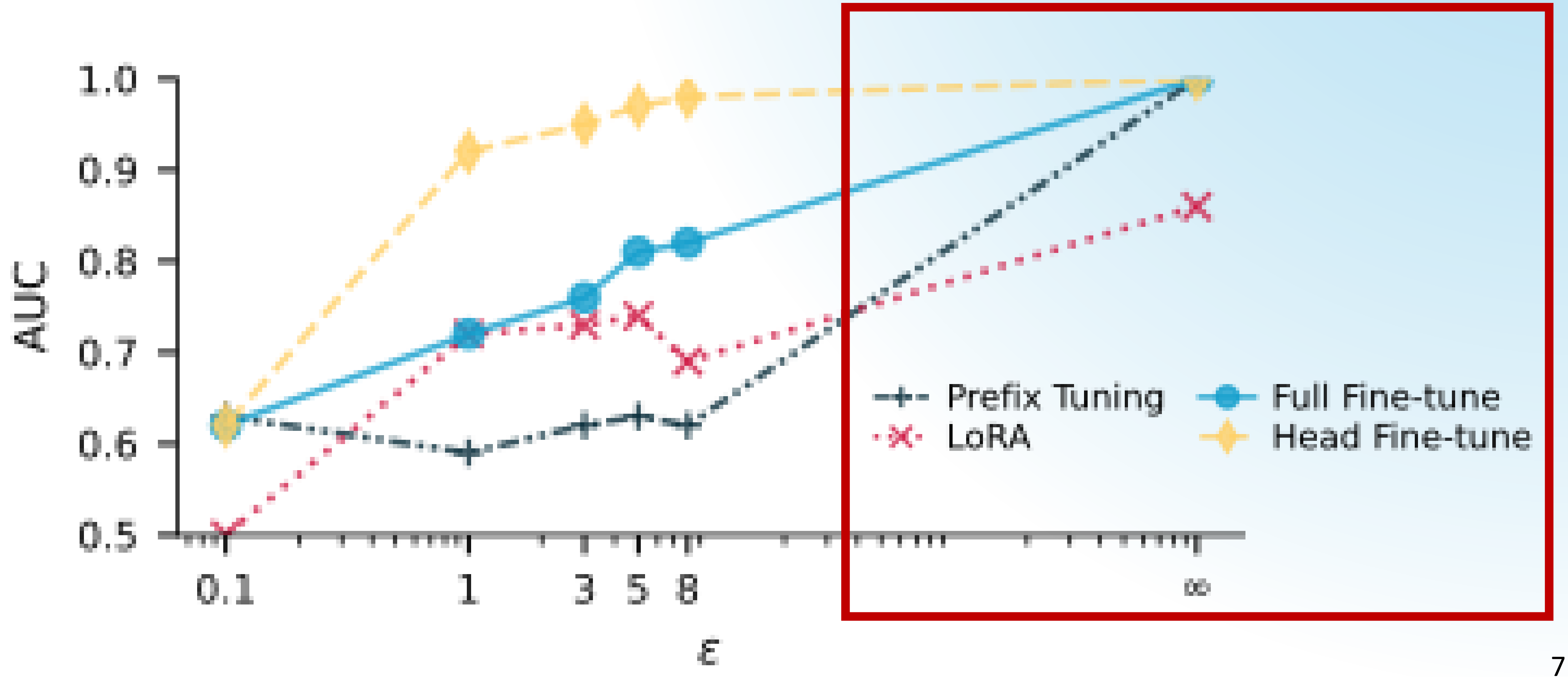

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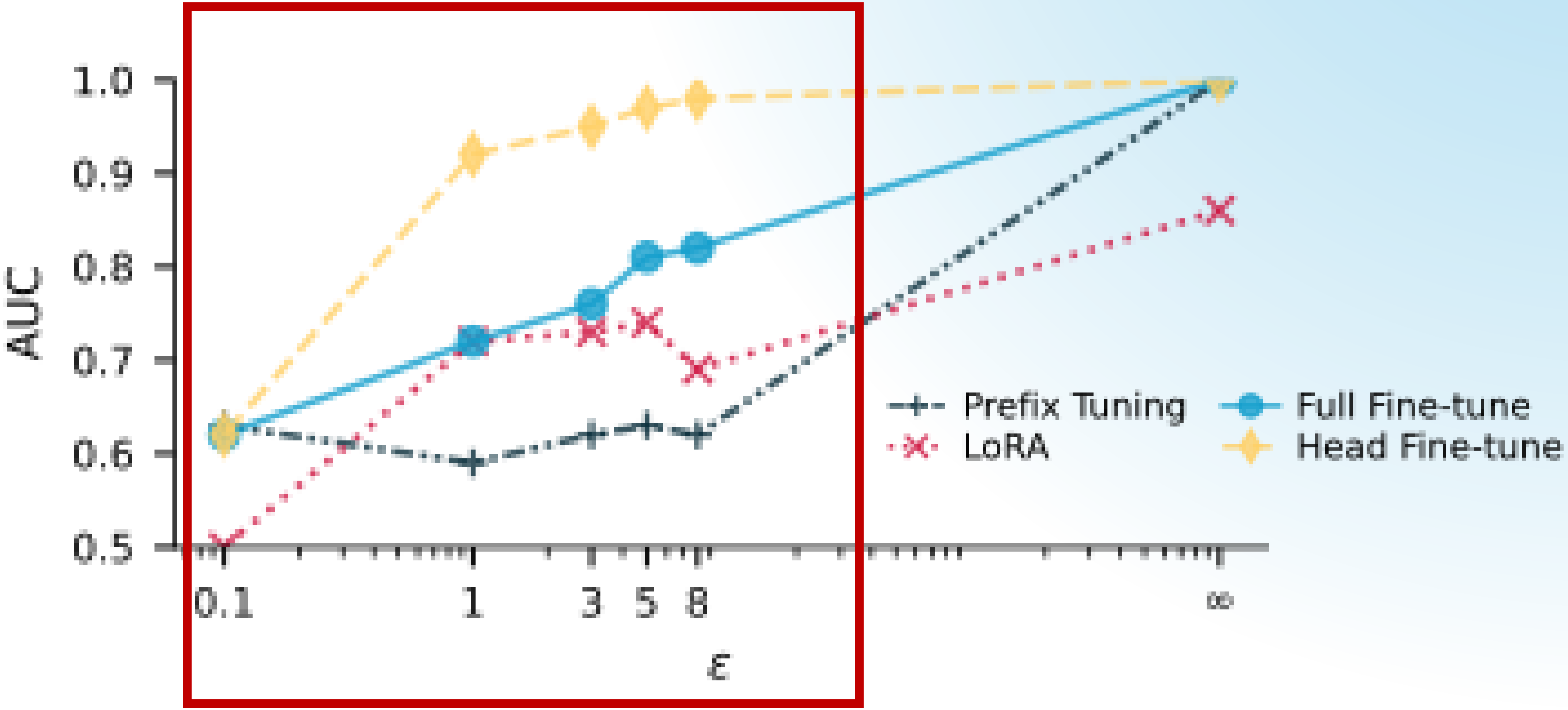
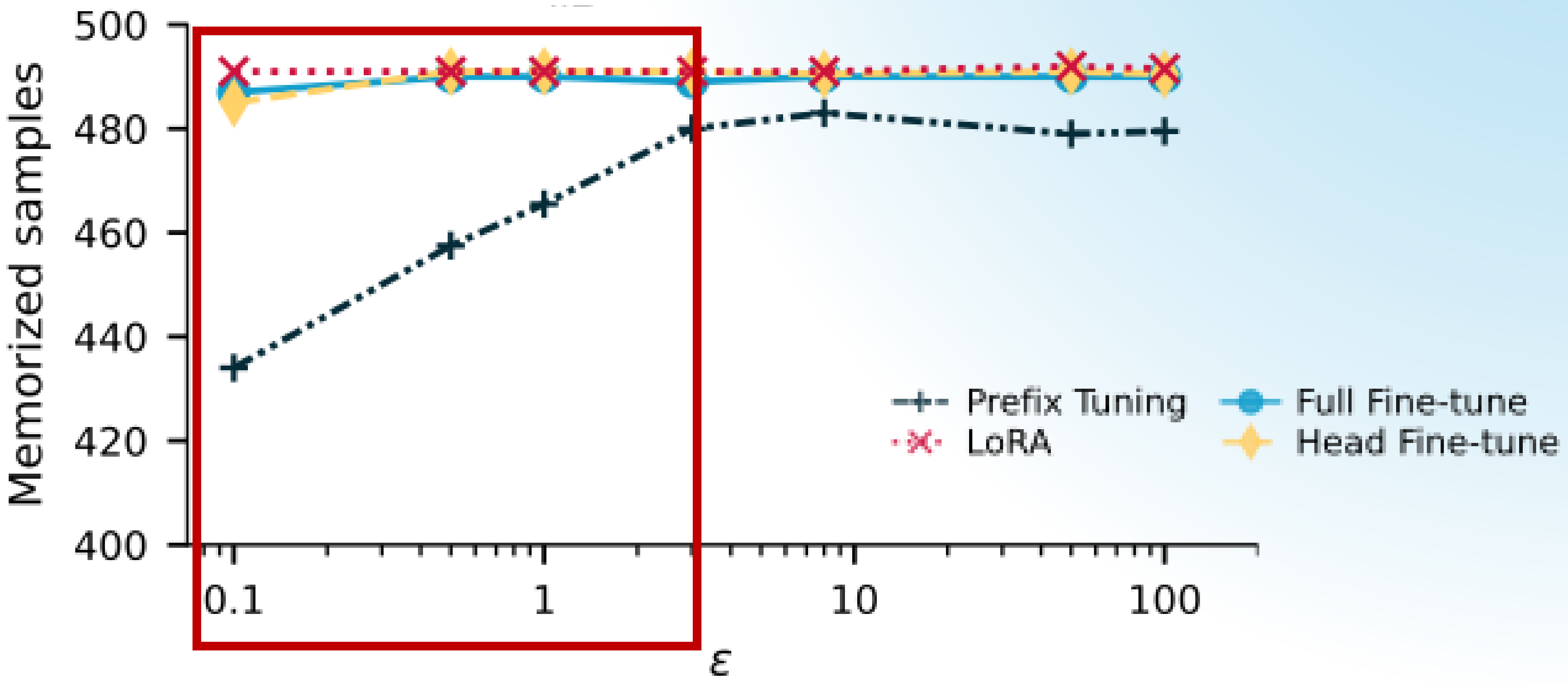


03 Adaptations Change Pretraining Leakage

DP adaptations influence the leakage of the pretraining data.



Our Vision: Holistic Privacy Audits

Formalize the audits of pretrain-adapt paradigm as adversarial games

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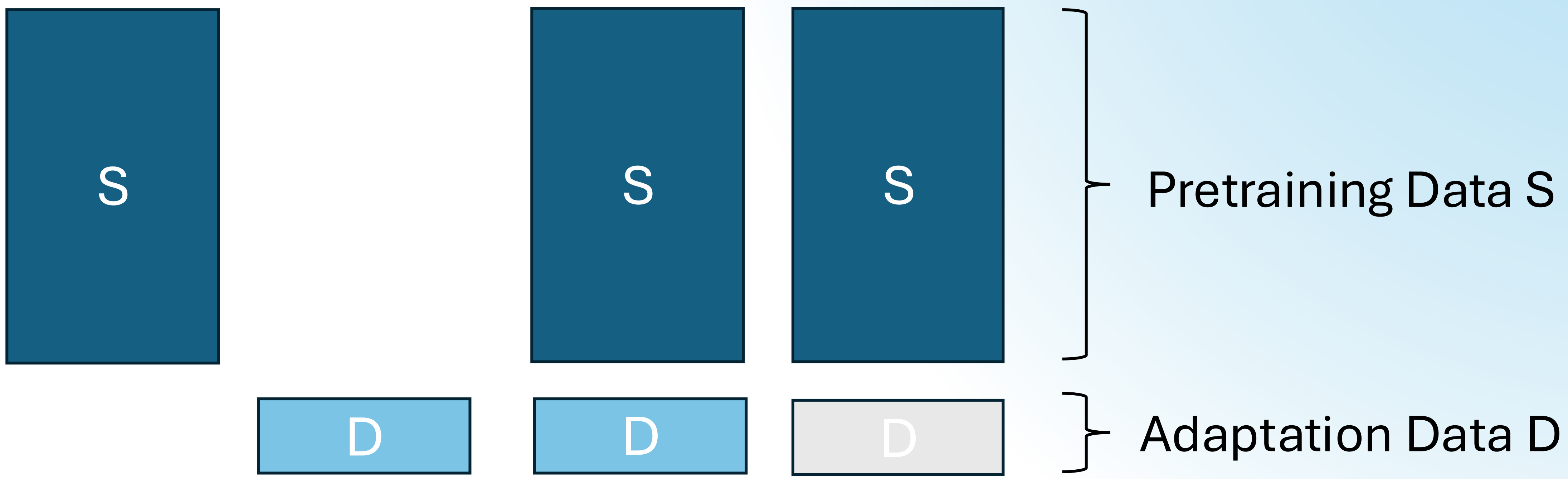

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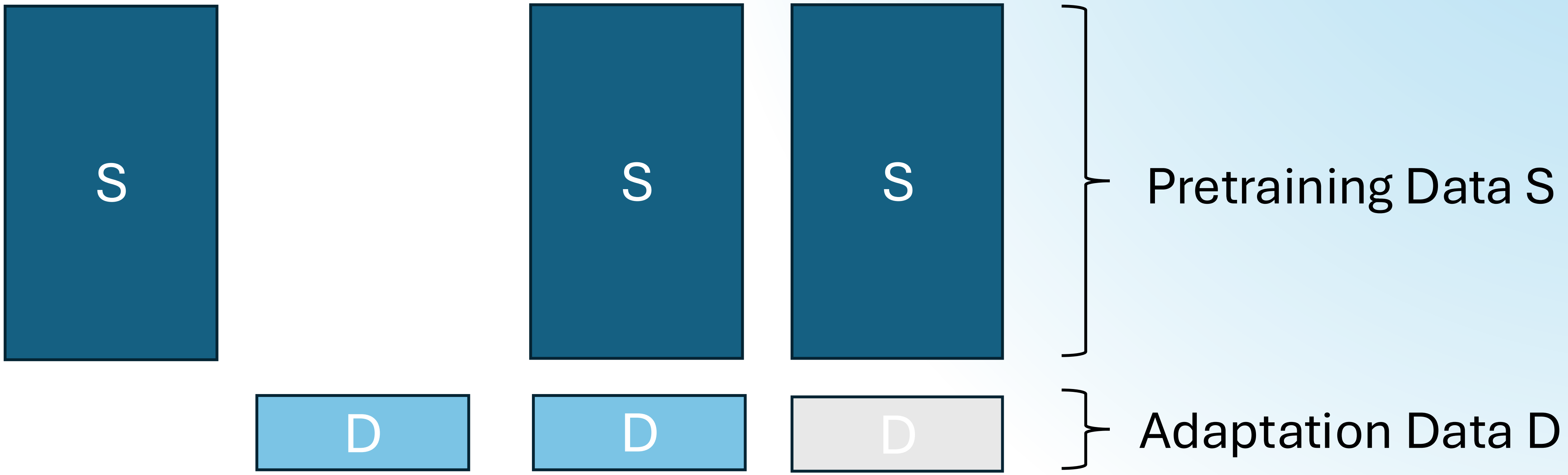

1. Stepping-stone for community on future privacy audits for foundation models
2. Guidance for practitioners on how to adapt the models with privacy by design

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Thank You!

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